ANDROID PROGRAMMING PRACTICAL 01

Module 1 · Hello World Android Application

Aim

Develop a simple Hello World Android application demonstrating project creation, Android Studio interface, Android components, AVD creation, and application execution.

Project name: HelloWorld

Requirements

Android Studio, Java, Android SDK, and an Android Virtual Device or compatible Android phone. Use the package name generated by your project and adjust imports/dependencies to the installed Android Studio template.

Procedure

1. Create a new Android Studio project using Empty Views Activity; set project name to HelloWorld and choose Java.

2. Inspect the Project, app, manifests, java, res, and Gradle Scripts sections.

3. In MainActivity.java, keep onCreate(), call super.onCreate(savedInstanceState), then setContentView(R.layout.activity_main).

4. In activity_main.xml, add a centered TextView with text “Hello, World!”, readable text size, and suitable padding.

5. Check AndroidManifest.xml for the launcher activity and exported setting required by the selected target SDK.

6. Open Device Manager, create or select an Android Virtual Device, start it, and press Run in Android Studio.

7. Confirm the app launches and displays Hello, World!; inspect Logcat if build or launch errors occur.

Implementation checklist

• Keep resource names lowercase and use unique view IDs.

• Make sure every referenced layout, menu, drawable, string, or Java class exists in the project.

• Build the project and resolve compile errors before running it.

• Use Logcat to diagnose runtime errors and permissions or manifest issues.

Expected result

The application launches in the emulator and displays the Hello World message.

Testing / observations

Record the device/API level, the actions tested, the observed output, and any errors corrected. Attach a screenshot of the successful run to your practical record if required by your institution.

Conclusion

The stated Android concept was implemented and verified through the steps and tests above.